

Average Velocity

$$\text{Average Velocity} = \frac{\text{Distance Traveled}}{\text{Time to Travel that Distance}} = \frac{\text{Change in Position}}{\text{Change in Time}} = \frac{\Delta \text{Position}}{\Delta \text{Time}}$$

Ex: Suppose you travel 30 miles in a half an hour.

$$\text{Average Velocity} = \frac{30 \text{ mi}}{0.5 \text{ h}} = 60 \text{ mph}$$

It is important to understand that this does not necessarily imply that you drove 60mph the whole way. In reality, there were certain points where you drove above 60mph and certain points where you drove below 60mph. This is just your average speed.

Question: Did you drive exactly 60mph at at least one time during your trip?

Answer: Intuition should tell you YES since you cannot drive both above and below 60mph without crossing through 60mph at some point.

Suppose we know the position of an object, O , as a function of time. Lets call it $s(t)$. Therefore, $s(t)$ represents the position (in meters) of O after some time t (in seconds).

Mathematical Expression	English Translation
$s(1)$	The position of O after 1s
$s(0)$	The initial position of O
$s(6)$	The position of O after 6s
$s(7) - s(4)$	O 's change position (distance traveled) between 4s and 7s
$s(3) - s(0)$	O 's change position (distance traveled) in the first 3s

If position, $s(t)$, of an object is known then the average velocity, $\bar{v}$, of between time $t = a$ and $t = b$ is given by:

$$\bar{v} = \frac{s(b) - s(a)}{b - a}$$

Ex: Consider a rock that is launched vertically upward. Let $s(t) = -16t^2 + 96t$ be the position (feet) of the rock after t seconds.

(a) What is the position of the rock after 1 s?

$$s(1) = -16(1)^2 + 96(1) = -16 + 96 = 80 \text{ ft}$$

(b) What was average velocity of the rock between 1 s and 3 s?

$$\bar{v}_{1 \rightarrow 3} = \frac{s(3) - s(1)}{3 - 1} = \frac{[-16(3)^2 + 96(3)] - [-16(1)^2 + 96(1)]}{2} = \frac{144 - 80}{2} = \frac{64}{2} = 32 \text{ ft/s}$$

(c) What was average velocity of the rock between 1 s and 2 s?

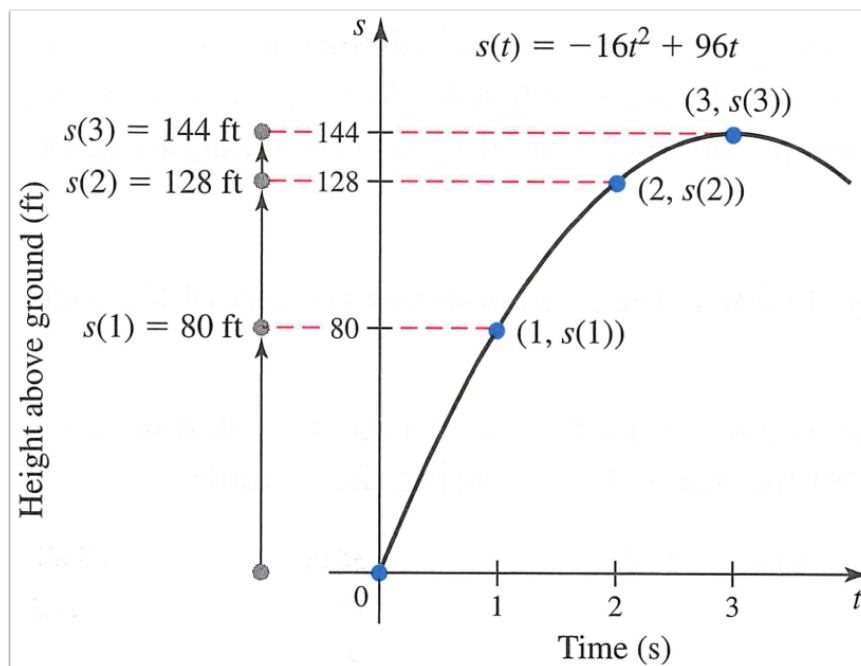
$$\bar{v}_{1 \rightarrow 2} = \frac{s(2) - s(1)}{2 - 1} = \frac{[-16(2)^2 + 96(2)] - [-16(1)^2 + 96(1)]}{1} = \frac{128 - 80}{1} = 48 \text{ ft/s}$$

(d) What was average velocity of the rock between 1 s and k s? (assume $k > 1$)

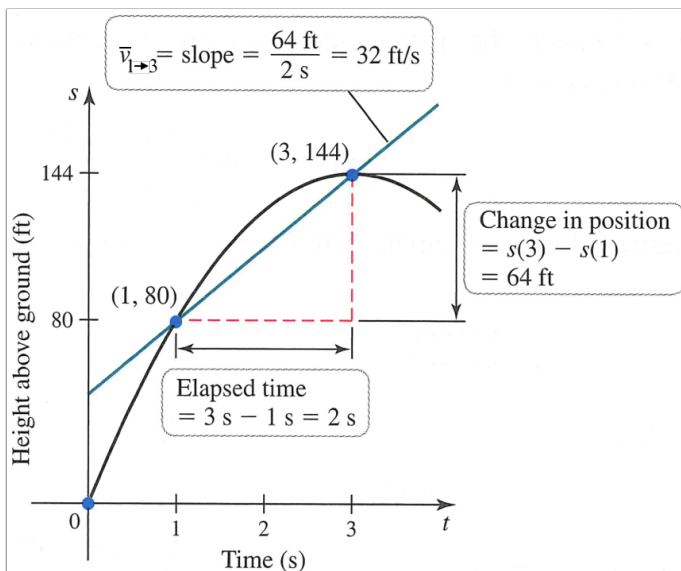
$$\bar{v}_{1 \rightarrow k} = \frac{s(k) - s(1)}{k - 1} = \frac{-16k^2 + 96k - 80}{k - 1} \text{ ft/s}$$

Now, lets see what this looks like graphically...

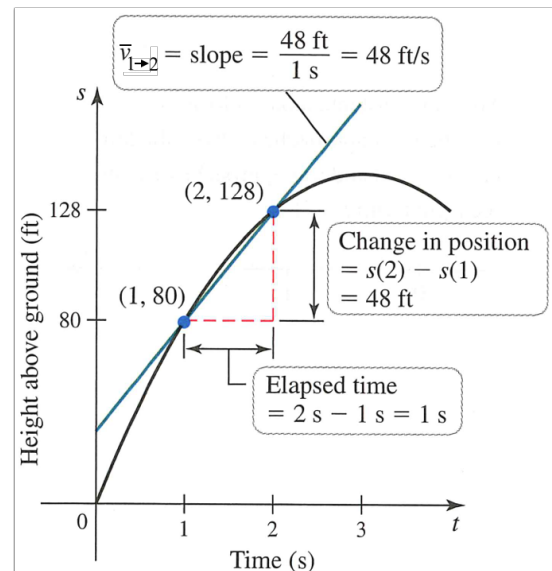
Position (height) of rock @ time $t = 1, 2, \& 3$



Average Velocity for 2 different time intervals $[1, 3]$ & $[1, 2]$



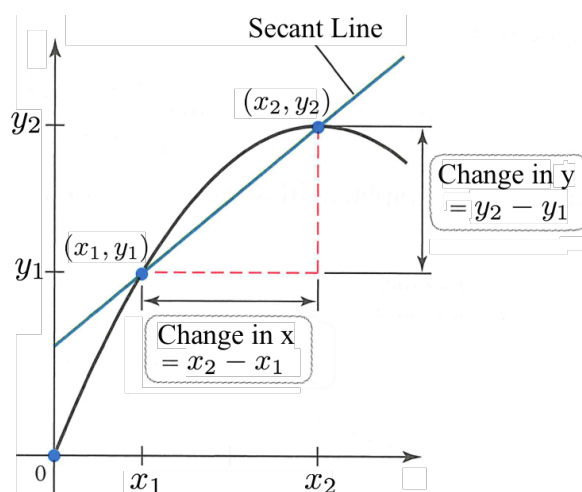
time interval: $[1, 3]$



time interval: $[1, 2]$

Secant Lines

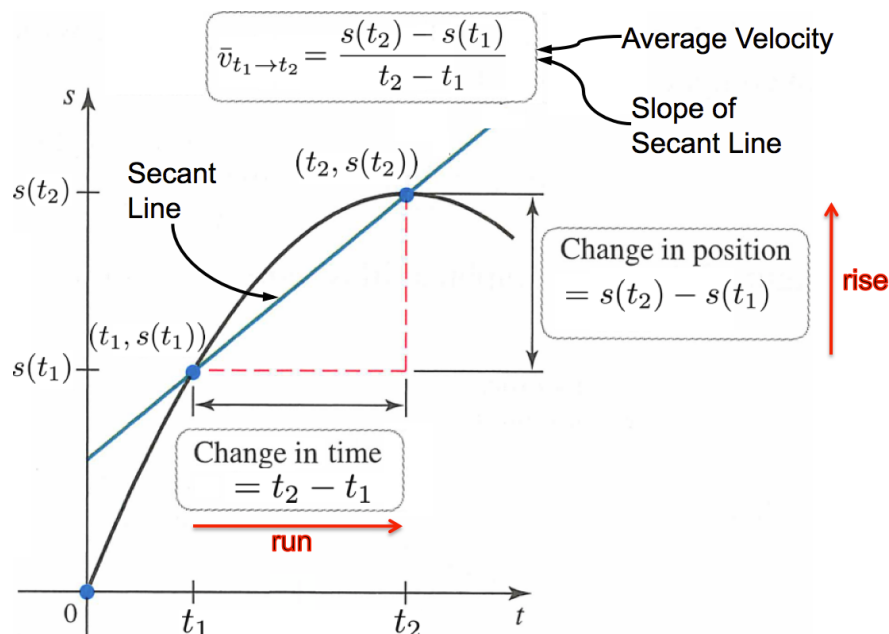
Defn: Let (x_1, y_1) and (x_2, y_2) be any two points on a curve. The line passing through (x_1, y_1) and (x_2, y_2) is called a **secant line**.



The slope of the secant line is

$$m_s = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Concept: The slope of any secant line of a position curve is an average velocity



Ex: Suppose $(2, -2)$ and $(6, 1)$ are two points on the graph of some function $f(x)$.

(a) Find the slope (m_S) of the secant line (S) between these two points

$$m_S = \frac{1 - (-2)}{6 - 2} = \frac{3}{4}$$

(b) Under what condition could this slope above be considered an average velocity?

If $f(x)$ was a position function of x (time)

Ex: Suppose the initial position of object was 5ft and its average velocity was 4ft/s during the first 8s. Find the equation of the secant line of $s(t)$ over the time interval $[0, 8]$ (secs).

To find the equation of any line, we need the slope, m , and a point, (x_1, y_1) , on the line which we plug into: $y - y_1 = m(x - x_1)$. Turns out that we are given both:

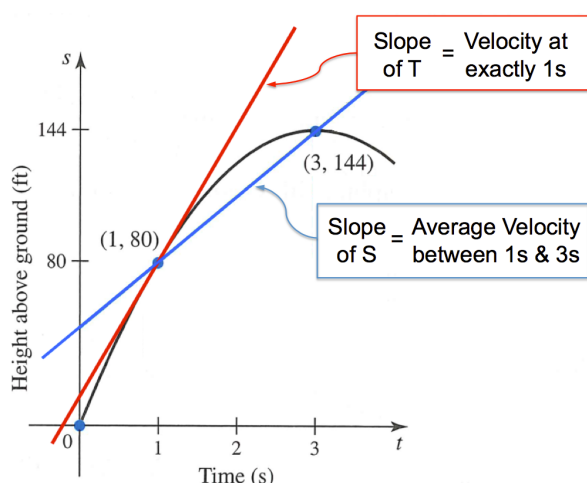
Since the average velocity = slope of the secant line, then $m = 4$.

Since the initial position is 5ft, then $s(0) = 5$ or $(x_1, y_1) = (0, 5)$.

Therefore, the equation of the secant line is: $y - 5 = 4(x - 0)$ or $y = 4x + 5$

Instantaneous Velocity (or just Velocity)

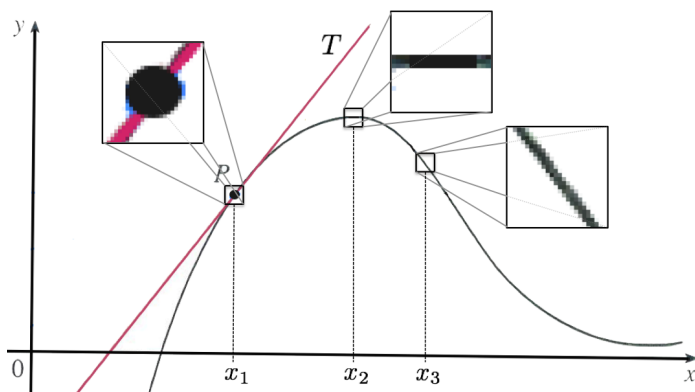
We now know that the average velocity can be interpreted as the slope of some secant line. But what about the exact velocity at a point? It turns out that the velocity is found from the slope of the tangent line.



Note: It is important to understand that it is not tangent or secant lines themselves that we are interested in, but rather the slope of these lines.

Tangent Lines

To get an idea of what a tangent line is, imagine zooming into one spot of a graph as shown below. Notice that in each box the graph appears to be a line. If you extend the line you see in the box near P , you will obtain the tangent line of the function at P .



Important Observations:

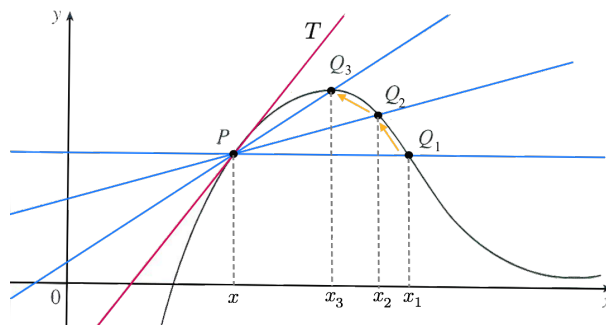
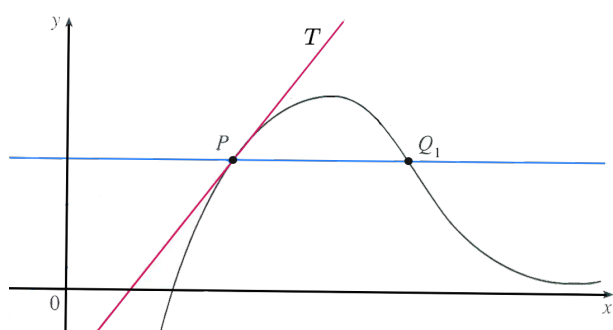
- ▶ At each x value, the graph has a different tangent line.
- ▶ Each tangent line has a slope, which measures how steep the line is.
- ▶ At any point on the graph (e.g. P), the function and the tangent line share the same steepness.

Therefore...

Concept: The slope of the tangent line at P tells us the steepness of the function at P .

In general, we need 2 points on a line to find the slope of the line. However, tangent lines only cross through 1 point, P . It turns out that the slope of the tangent line can be approximated from the the slope of a secant line that is very close to the tangent line.

Consider adding a point Q_1 to the curve and draw a secant line through P and Q_1 . We can compute the slope of this secant line, but clearly, this won't give us the slope of the tangent line. Now, imagine that we added more points $Q_2, Q_3, \dots$ closer to P . What will happen to the blue secant lines as the points keep getting closer to P ?



It should be clear that the secant lines will eventually be very close to the tangent line. Thus, the slopes of secant lines will also eventually be very close to the slope of the tangent line.

Ex: Given the position function, $s(t) = 5e^t$ (meters), estimate the instantaneous velocity at $t = 2.5$ s to within 2 decimal places.

In order to estimate the velocity(slope of the tangent line) at $t = 2.5$ s we need to use a sequence of average velocities (slopes of secant lines) near $t = 2.5$ s (just like above). Lets compute the average velocities on the following intervals:

$[2.5, 2.51]$	$\frac{s(2.51) - s(2.5)}{2.51 - 2.5} = \frac{5e^{2(2.51)} - 5e^{2(2.5)}}{0.01} = 61.2181$
$[2.5, 2.51]$	$\frac{s(2.501) - s(2.5)}{2.501 - 2.5} = \frac{5e^{2(2.501)} - 5e^{2(2.5)}}{0.001} = 60.9429$
$[2.5, 2.5001]$	$\frac{s(2.5001) - s(2.5)}{2.5001 - 2.5} = \frac{5e^{2(2.5001)} - 5e^{2(2.5)}}{0.0001} = \underline{60.9155}$
$[2.5, 2.50001]$	$\frac{s(2.50001) - s(2.5)}{2.50001 - 2.5} = \frac{5e^{2(2.50001)} - 5e^{2(2.5)}}{0.00001} = \underline{60.9128}$

Notice that the average velocities appear to be converging some fixed value as the intervals continue to shrink. The value that we would eventually reach is the instantaneous velocity at $t = 2.5$ s. For this example, we just estimate the this value as 60.91m/s